

pfSense Firewall Lab

1. What I did!

In this lab, I will create a small virtual network with three machines:

- A firewall (pfSense) that controls traffic
- An attacker machine (Kali Linux)
- A victim machine (Ubuntu)

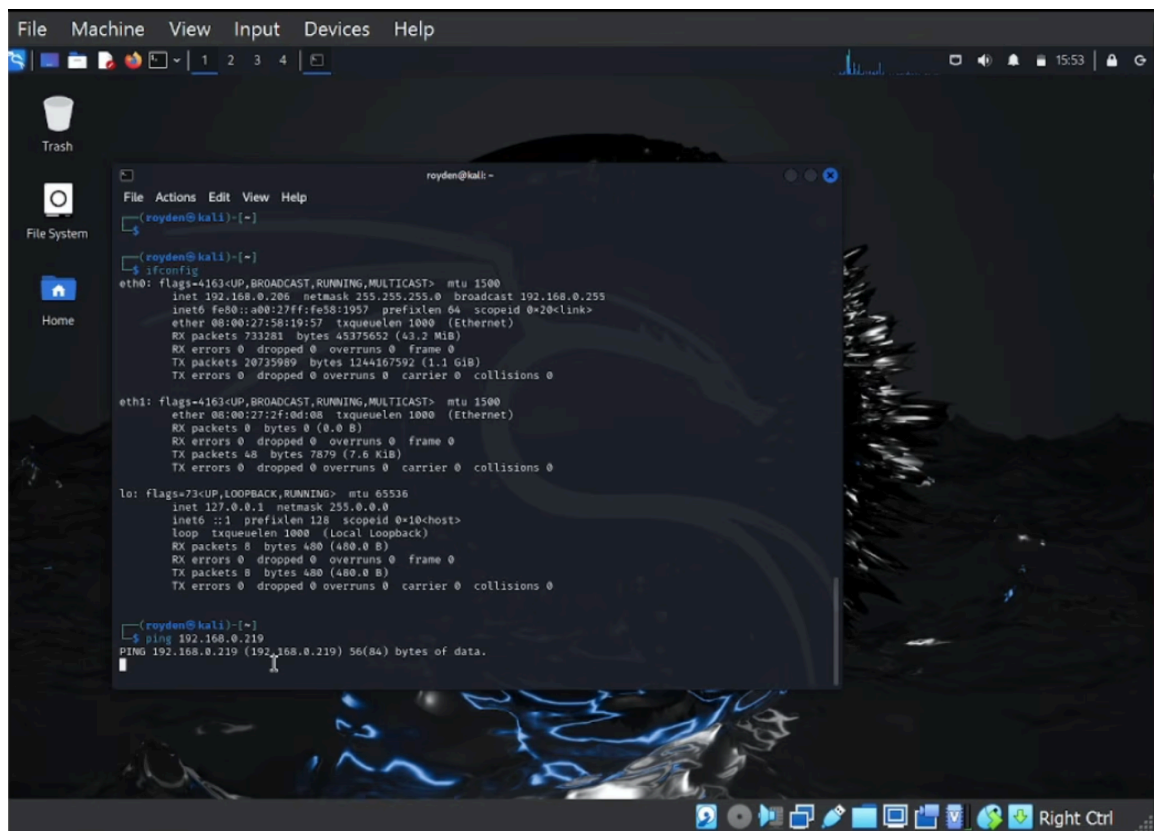
I will send a fake attack(DoS) from the attacker and then block it using the firewall.

2. Basic Setup

I turned on all the machines.

pfSense will automatically give the Ubuntu machine an IP address.

I made sure Kali and Ubuntu can communicate (ping test).

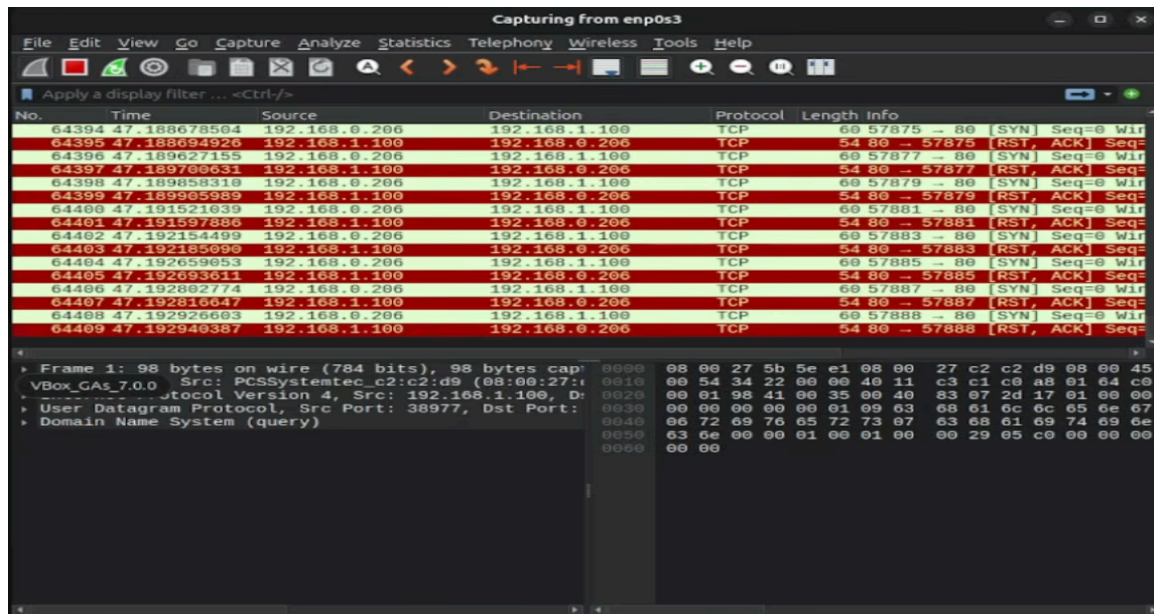


3. Simulate an Attack

On Ubuntu, I opened Wireshark to watch network traffic.

On Kali, I ran a command (hping3) to send a large amount of traffic to Ubuntu.

You should see a spike in traffic on Ubuntu.



4. Block the Attack

I went to the pSense web interface.

I created a rule to block traffic coming from the Kali machine.

I applied the rule

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐		0/38.12 MiB	IPV4 TCP	192.168.0.206	*	192.168.1.100	*	*	none		

I checked my Kali machine and ran the command again

```
royden@kali: ~  
File Actions Edit View Help  
64 bytes from 192.168.1.100: icmp_seq=82 ttl=63 time=0.962 ms  
^C  
— 192.168.1.100 ping statistics —  
82 packets transmitted, 13 received, 84.1463% packet loss, time 82756ms  
rtt min/avg/max/mdev = 0.710/0.982/1.307/0.194 ms  
  
(royden@kali)-[~]  
$ sudo hping3 -S -p 80 --flood 192.168.1.100  
HPING 192.168.1.100 (eth0 192.168.1.100): S set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown  
^C  
— 192.168.1.100 hping statistic —  
522773 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$  
  
(royden@kali)-[~]  
$ sudo hping3 -S -p 80 --flood 192.168.1.100  
HPING 192.168.1.100 (eth0 192.168.1.100): S set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown
```

The traffic should stop immediately.

```
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help  
Apply a display filter ... <Ctrl-/>  
No. Time Source Destination Protocol Length Info  
1 0.000000000 192.168.1.100 192.168.0.217 TCP 74 35368 → 8080 [SYN] Seq=0 Win  
2 1.012956264 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
3 2.037896576 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
4 3.060949708 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
5 4.085684708 192.168.1.100 192.168.0.217 TCP 74 [TCP Retransmission] 35368  
6 5.300916196 PCSSystemtec_c2:c2: PCSSystemtec_5b:5e: ARP 42 Who has 192.168.1.1? Ielt 1  
7 5.301613298 PCSSystemtec_5b:5e: PCSSystemtec_c2:c2: ARP 60 192.168.1.1 is at 08:00:27:  
8 27.393327626 192.168.1.100 192.168.1.1 DHCP 329 DHCP Request - Transaction  
9 27.393867667 192.168.1.1 192.168.1.100 DHCP 342 DHCP ACK - Transaction  
  
Frame 1: 74 bytes on wire (592 bits), 74 bytes captured on interface eth0  
Ethernet II, Src: PCSSystemtec_c2:c2:d9 (08:00:27:00:00:00), Dst: 08:00:27:00:00:00  
Internet Protocol Version 4, Src: 192.168.1.100, Destination: 192.168.0.217  
Transmission Control Protocol, Src Port: 35368, Destination Port: 8080  
Sequence Number: 0, Window Size: 0, Length: 40  
Flags: S, Seq=0, Win=0, Len=0  
0000 08 00 27 5b 5e e1 08 00 27 c2 c2 d9 08 00 45  
0010 00 3c d6 1f 40 00 40 06 e1 0e c0 a8 01 64 c0  
0020 00 d9 8a 28 1f 90 6b 08 46 20 00 00 00 00 a0  
0030 fa f0 83 bc 00 00 02 04 05 b4 04 02 08 0a f2  
0040 29 97 00 00 00 00 01 03 03 07
```